Nathaniel Murcin

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Education

Georgia Institute of Technology GPA: 4.00/4.00

B.S. Aerospace Engineering | *B.S./M.S. Aerospace Engineering* *B.S. Expected May 2027*

Minor in Astrophysics *M.S. Expected May 2028*

- Coursework: Aerothermodynamics, Aerodynamics, Fluid Dynamics, Jet & Rocket Propulsion, Structural Analysis

Experience

Blue Origin Cape Canaveral, FL

Fluids & Manufacturing Engineering Intern *May 2026 – Aug. 2026*

Cryogenic Tank Farm Design

- Designed a cryogenic tank-farm P&ID for a critical-path ground-test facility, reducing component count $\sim$60% versus heritage designs by downsizing, eliminating branches, and challenging requirements.
- Sized and selected cryogenic piping, valves, and relief devices using Cv, Darcy-Weisbach, line-sizing, and overpressure-relief analyses, then developed the BOM and coordinated suppliers.
- Performed preliminary hazard analysis for a major cryogenic fluid system, evaluating component failure modes and consequences against safeguards while documenting CONOPS for formal HAZOP review.

Lunar Engineering

- Analyzed a load-bearing lunar hardware fixture to flight-structure standards via hand calculations and Creo FEA (torque, bolt shear/tension, preload, slip, tear-out, bending, CTE) against yield/ultimate margins; passed PDR.
- Iterated through 6 Creo design revisions, cutting fastener count from 8 to 1 while preventing part slippage and damage.

Georgia Tech Experimental Rocketry (GTXR) Atlanta, GA

Fin Design & Manufacturing Lead *Aug. 2023 – Present*

- Led fin architecture and core-material trade study (flutter margin, mass, aero performance, repeatability); selected a hexagonal short-taper G10 design, accepting +1.24 lb for flutter margin and consistency across 8 flight fins.
- Led design of a modular four-tower fin-alignment fixture in SolidWorks after a 3D scan found a 1.04° clocking failure; with a standardized attachment process, cut worst-case error 73% to 0.277° and held cant within 0.1°.
- Designed plate-twist test apparatus and tested composite sandwich panels on an Instron to measure in-plane shear modulus, validating the ANSYS ACP model within 5% (mean) for fin flutter predictions.
- Designed and fabricated a flight-critical NPT pressure-boundary passthrough with the fin team in 14 hours after a flight-unit failure; passed hydrostatic proof at 2$\times$ MEOP and enabled second-stage ignition.
- Recommended an I-beam rail and larger base for the V2 mock launch rail with the structures lead; aided design and load-path analysis with bending and Euler buckling checks.

Pursuit Aerospace Newburyport, MA

Engineering Intern *Aug. 2025 – Dec. 2025*

- Designed and deployed 15 quick-change marking fixtures using GD&T and DFM, cutting load/unload time 83%.
- Designed Go/No-Go gauges that reduced inspection time by 95%.
- Developed an in-house precision cleaning process for a Rolls-Royce component, qualified for production, cutting lead time by 3 days and saving $1,000/part.

Ben T. Zinn Combustion Laboratory Atlanta, GA

Undergraduate Researcher *Jan. 2026 – Present*

- Designed and installed regulated multi-gas manifold hardware distributing H_2, He, GOx, N_2, and natural gas to predetermined inlets on a reacting jets-in-crossflow test rig.
- Helped build the laser PIV system and processed PIV data in MATLAB; building hardware for a follow-on experiment.

Publications

- **Co-author & presenter:** *Viability of Thermoformed Closed-Cell Foam Cores for Carbon Fiber Sandwich Rocket Fins*, AIAA Regional Student Conference Proceedings, 2026 (AIAA 2026-112793).
- **Co-author & presenter:** *Shear Modulus Testing in Composite Sandwich Panels*, AIAA Regional Student Conference Proceedings, 2024 (AIAA 2024-85690).

Technical Skills

Design: P&ID design, line & relief sizing, GD&T, DFM, trade studies, composite structures
Analysis: Structural FEA, flutter, hazard analysis, Cv & pressure-drop analysis
Software: SolidWorks, Creo, MATLAB, Python
Test & Fabrication: Instron, PIV, 3D scanning & metrology, composite layups, CNC mill, lathe, waterjet, 3D printing